CS34020 / CSM4220 Assignment Short Summary

This document is not intended to replace the full assignment specification!

Deadline: 23 November 2026, **Worth**: 50% of the module

The task is to develop a **computer vision system to support a strawberry-picking robot**. You do **not** need to implement the robot itself. Your vision system should:

- identify strawberries and suitable stem picking locations in images/video;
- track the picking locations through video;
- indicate **"pick now"** on screen or at the command line when a strawberry has been identified and located for picking;
- avoid picking **unripe fruit** and, if possible, avoid overripe or mouldy fruit.

Your solution should use Python and OpenCV and must run on the CS laboratory machines without installing additional libraries. You may use an appropriate pre-trained model, but you should not train your own model.

Where the marks come from

1. Workshops: 10%. Submit your work and include answers from each workshop in your report.

2. Implementation: 20%. Develop the main vision system. The approach is your choice. You need to **test that it works correctly** and evaluate its quality using suitable **testing and statistics**. Higher marks require successful detection and tracking in video, including a good way of determining **when and which strawberry to pick**.

You may either develop the implementation yourself or make use of AI. If using AI, **document how you used it**, including the platform, prompts, skills and guardrails. You do not receive marks for code you did not produce yourself, but a **well-reasoned, evidenced and understood AI-assisted approach can receive marks**.

3. Flair / optional extras: 20%. First, get the basic system working. Then gain these marks by doing something particularly well or adding ambitious features. **Extras do not earn marks if the basic functionality is not mostly complete and working.**

4. Report: 20%. Maximum 8 pages, including images. It should contain: Introduction, Implementation, Self-evaluation, References and AI statement.

5. Presentation and group discussion: 30%. Present your solution to a small group (we will organise these) and explain its **advantages, disadvantages, testing and evaluation**. You will then compare the approaches and discuss which would be best for strawberry picking. Marks include your slides, critical analysis, **questions, answers and engagement in the discussion**. Your understanding of your own solution is important.

Submission

- **report** (max 8 pages) → Turnitin
- ZIP of your **implementation** and **presentation slides** (max 10 slides total) → Blackboard
- participate in the assessed **group discussion**.